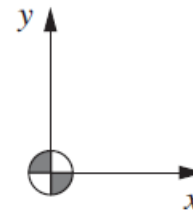
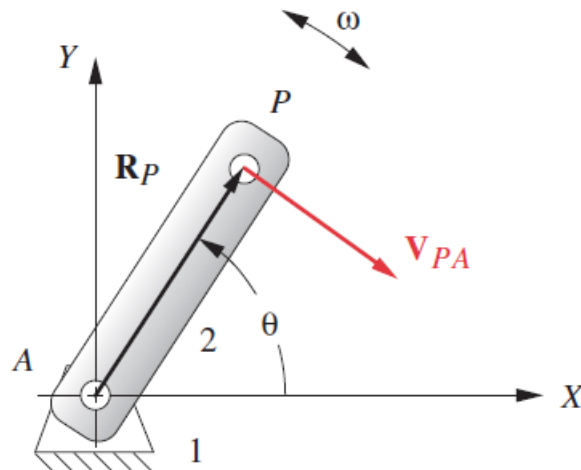


Linkage Analysis --Velocity (Ch. 6)

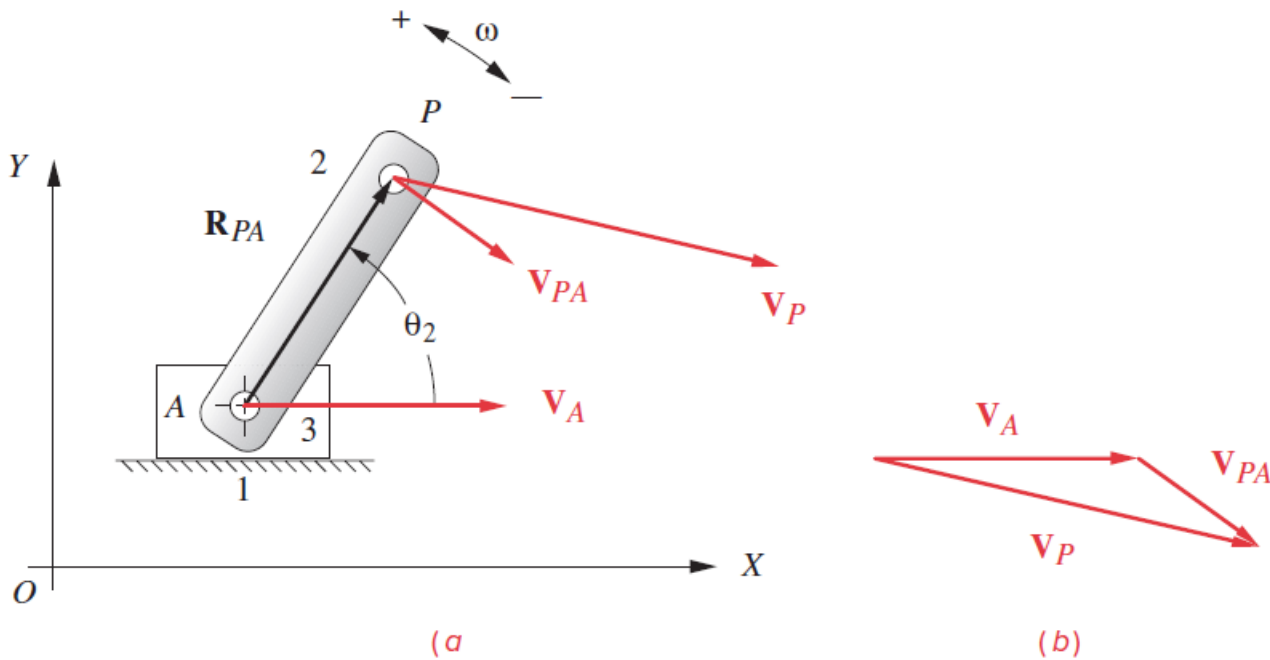
- Fundamentals
 - *Rotation*



$$\mathbf{R}_{PA} = p e^{j\theta}$$

$$\mathbf{V}_{PA} = \frac{d\mathbf{R}_{PA}}{dt} = p j e^{j\theta} \frac{d\theta}{dt} = p \omega j e^{j\theta}$$

– *Rotation and translation (relative velocity)*



$V_{PA} = V_P - V_A$ Difference between two absolute velocities (relative to the global coordinate frame XOY); *relative velocity with respect to the frame fixed at A and parallel to XOY .*

$$V_P = V_A + V_{PA} = V_A + p\omega je^{j\theta}$$

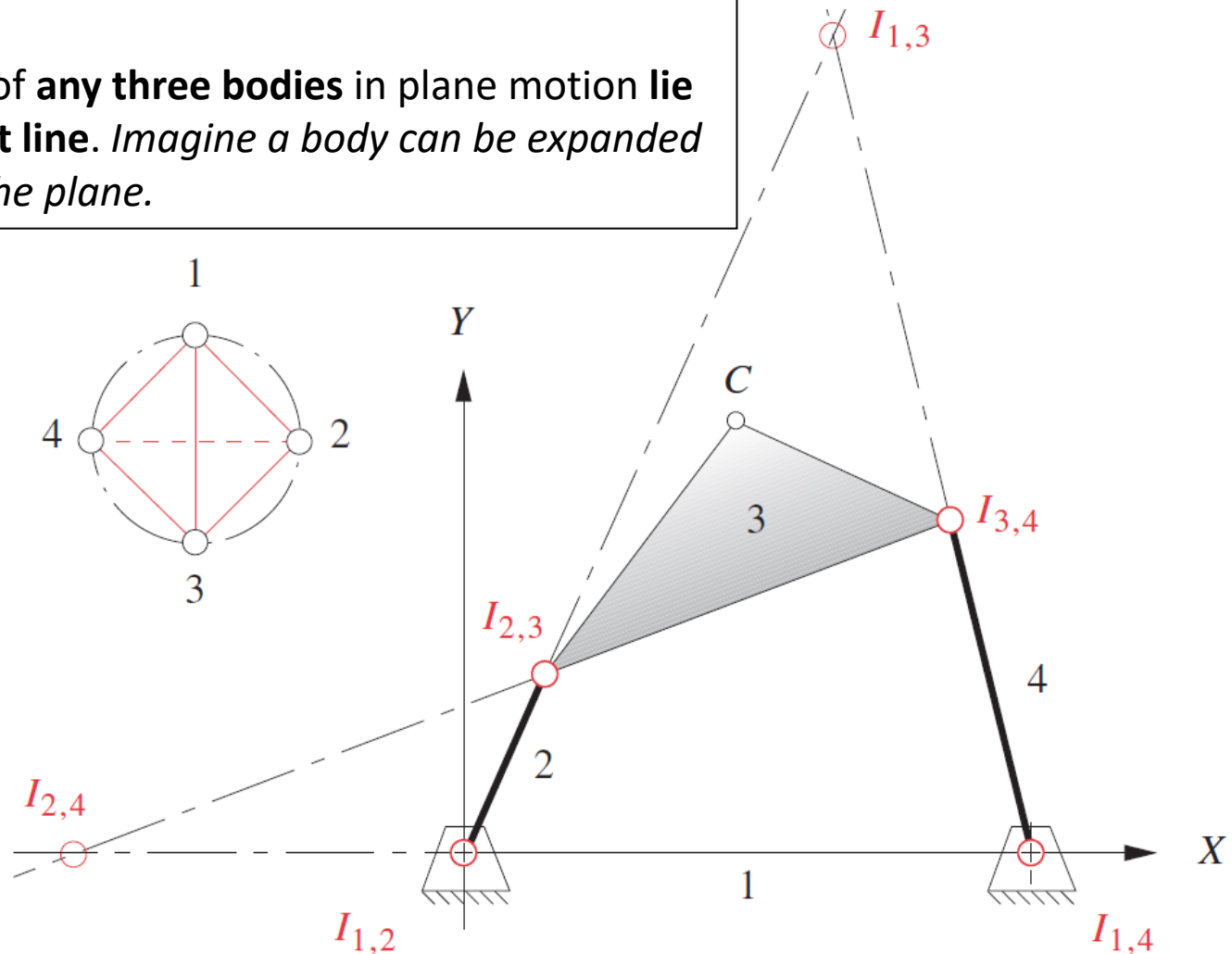
Velocity = translation component + rotation component

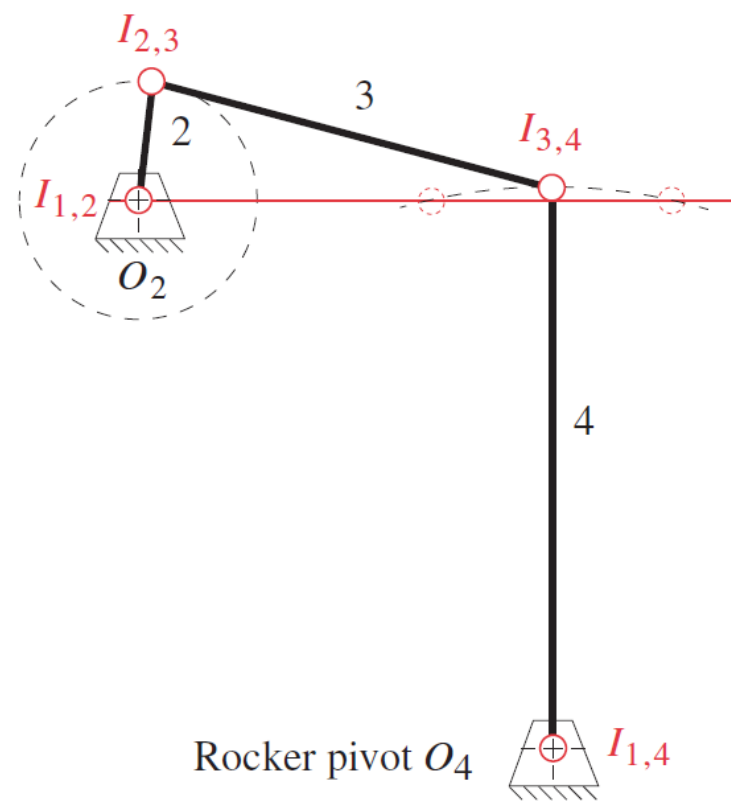
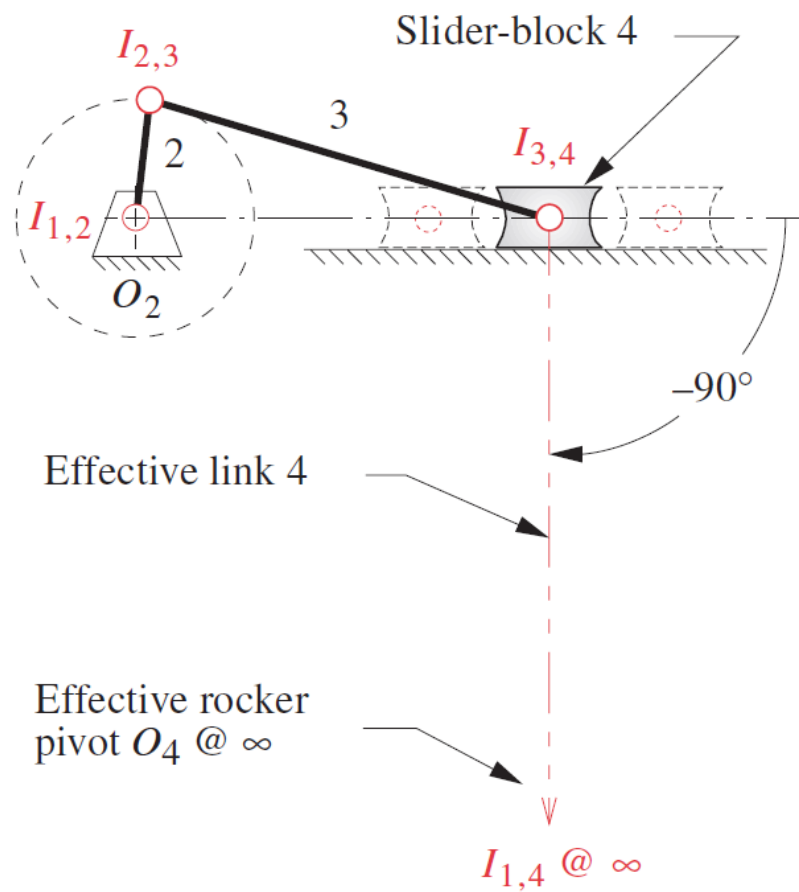
– *Instant Center of Velocity*

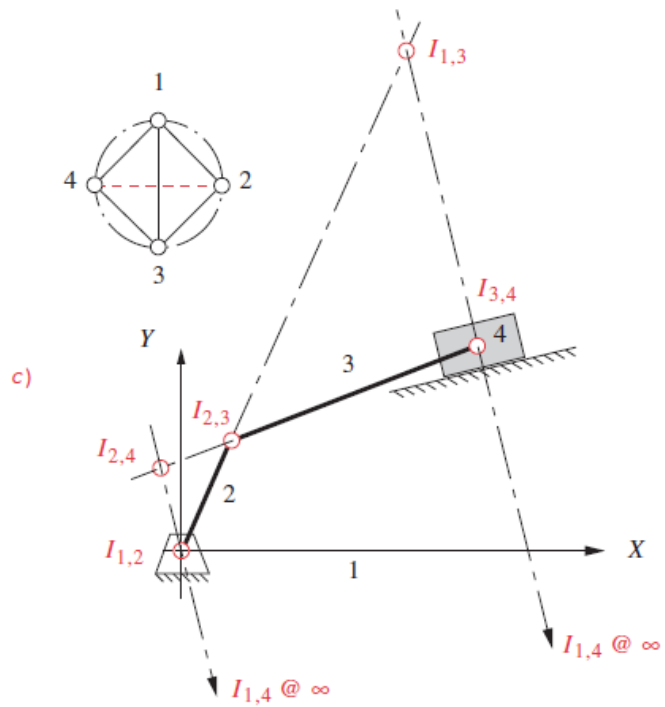
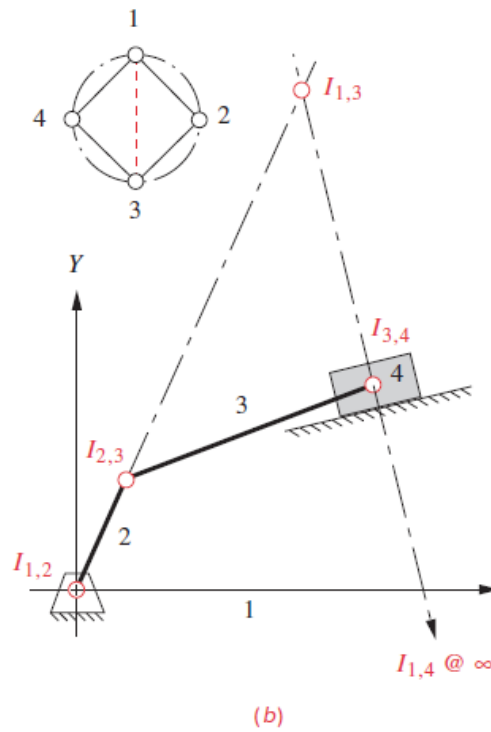
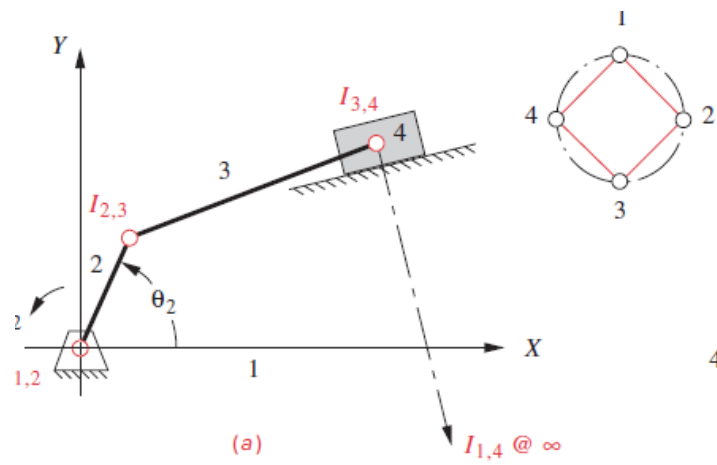
A point, **common to two bodies** in plane motion, which has the **same instantaneous velocity** in each body.

Kennedy's rule:

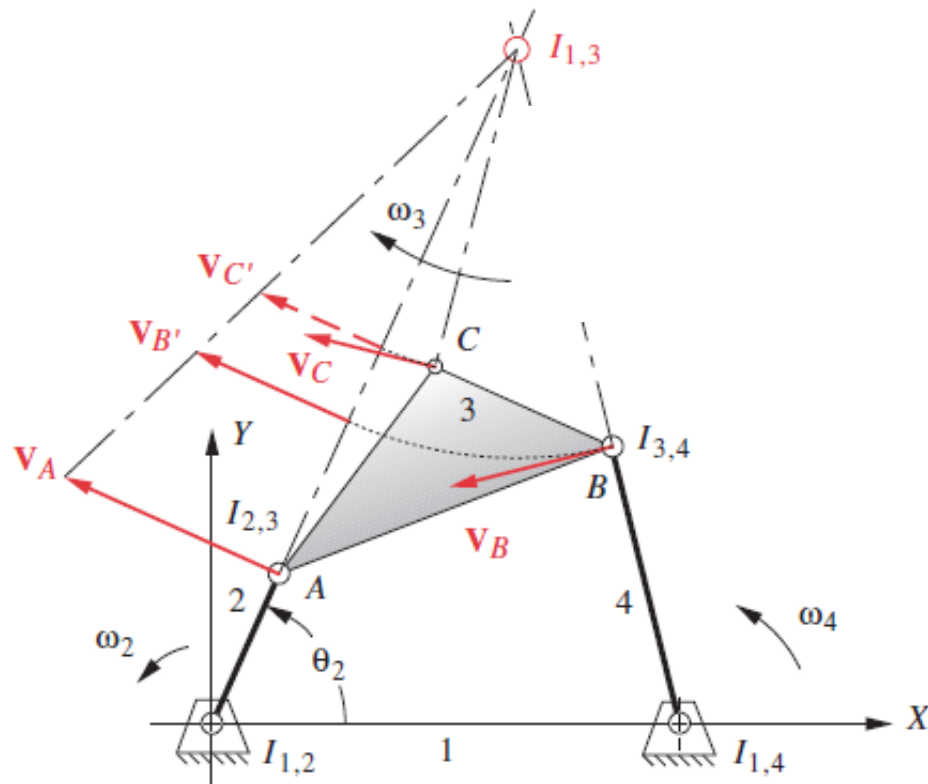
The instant centers of **any three bodies** in plane motion **lie on the same straight line**. *Imagine a body can be expanded infinitely in size on the plane.*







- **Velocity analysis when the linkage is at a position – graphic method (*mainly*)**
 - ***Applications of ICs (instant centers)***



$$\omega_3 = \frac{v_A}{(AI_{1,3})}$$

$$v_B = (BI_{1,3})\omega_3$$

$$\omega_4 = \frac{v_B}{(BO_4)}$$

$$v_C = (CI_{1,3})\omega_3$$

Directions are as shown and are determined from the ***circular motion around the IC***

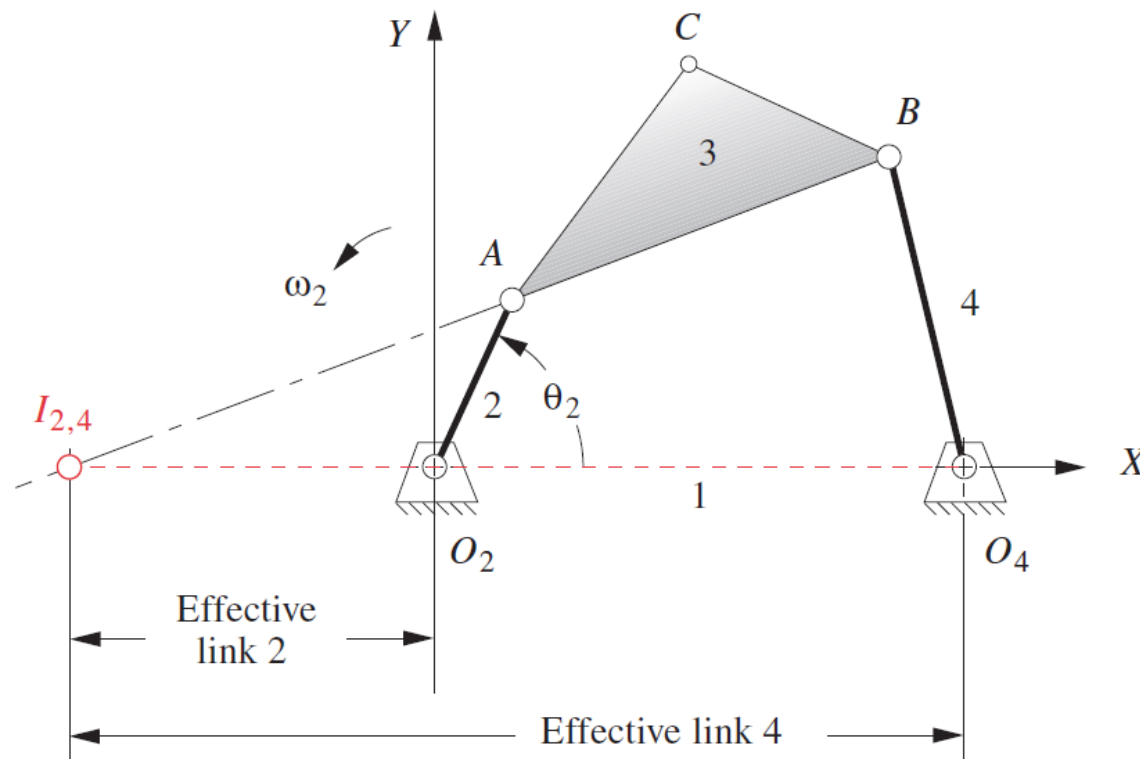
– **Velocity ratio** (between output velocity and input velocity) $m_V = \frac{\omega_4}{\omega_2}$

$$\omega_4 O_4 I_{2,4} = \omega_2 O_2 I_{2,4}$$

Speed of the IC $I_{2,4}$

$$m_V = \frac{\omega_4}{\omega_2} = \frac{O_2 I_{2,4}}{O_4 I_{2,4}}$$

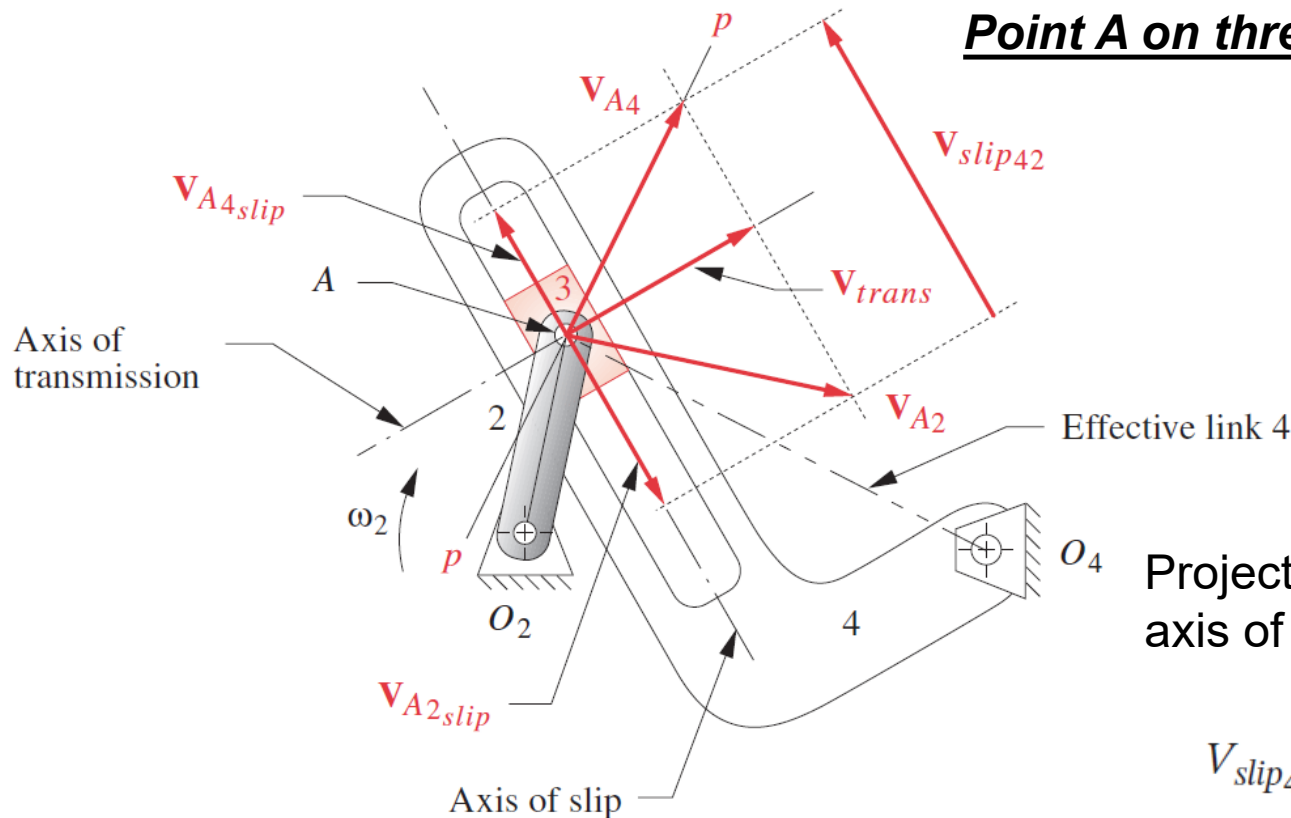
The sign depends on the directions of angular velocities



Effective link pairs: two lines, mutually parallel, drawn through the fixed pivots and intersecting the coupler extended.

– Velocity of slip (when there is a sliding joint)

Point A on three links: Links 2, 3, 4



$$v_{A_2} = (AO_2)\omega_2$$

$$v_{A_3} = v_{A_2}$$

$$V_{A_4} = AO_4 \omega_4$$

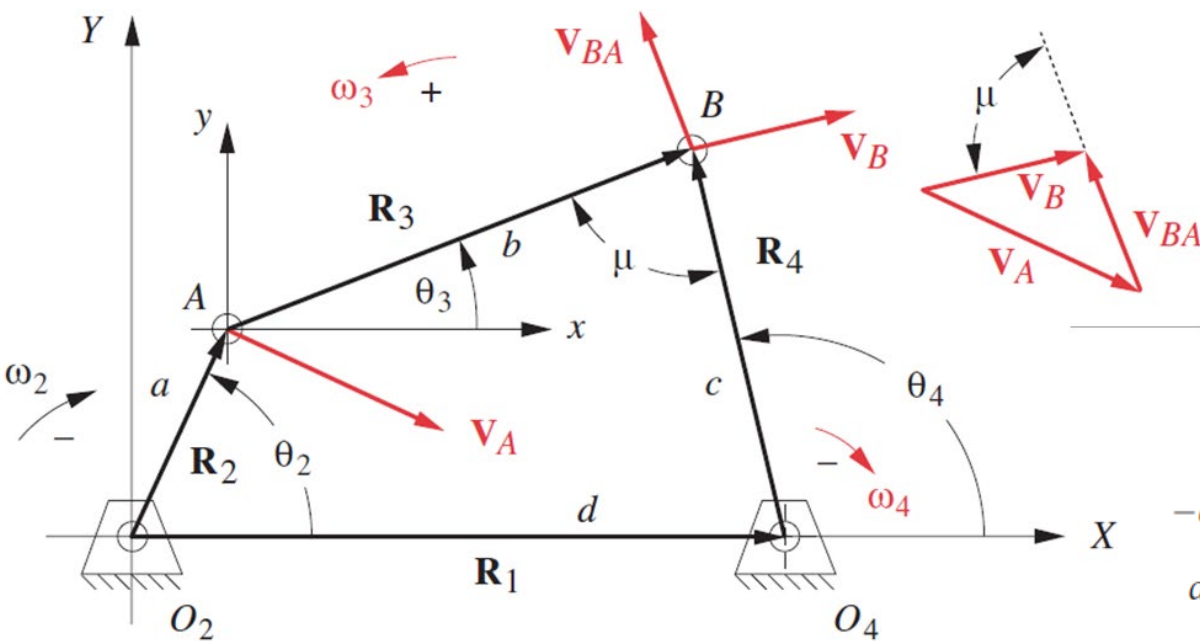
Project v_{A_2} and V_{A_4} onto the axis of slip to get V_{A_2slip} and V_{A_4slip}

$$V_{slip42} = V_{A_4slip} - V_{A_2slip}$$

$$\omega_4 = \omega_3 = \frac{V_{A_4}}{AO_4}$$

- Velocity analysis when the linkage is at any position – Four-bar linkage**

$$R_2 + R_3 - R_4 - R_1 = 0 \iff ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - d = 0$$



Differentiate the above with time and note

$$\frac{d\theta_2}{dt} = \omega_2; \quad \frac{d\theta_3}{dt} = \omega_3; \quad \frac{d\theta_4}{dt} = \omega_4$$

$$ja\omega_2 e^{j\theta_2} + jb\omega_3 e^{j\theta_3} - jc\omega_4 e^{j\theta_4} = 0$$

$$\underline{V_A} + \underline{V_{BA}} - \underline{V_B} = 0$$

$$-a\omega_2 \sin\theta_2 - b\omega_3 \sin\theta_3 + c\omega_4 \sin\theta_4 = 0$$

$$a\omega_2 \cos\theta_2 + b\omega_3 \cos\theta_3 - c\omega_4 \cos\theta_4 = 0$$

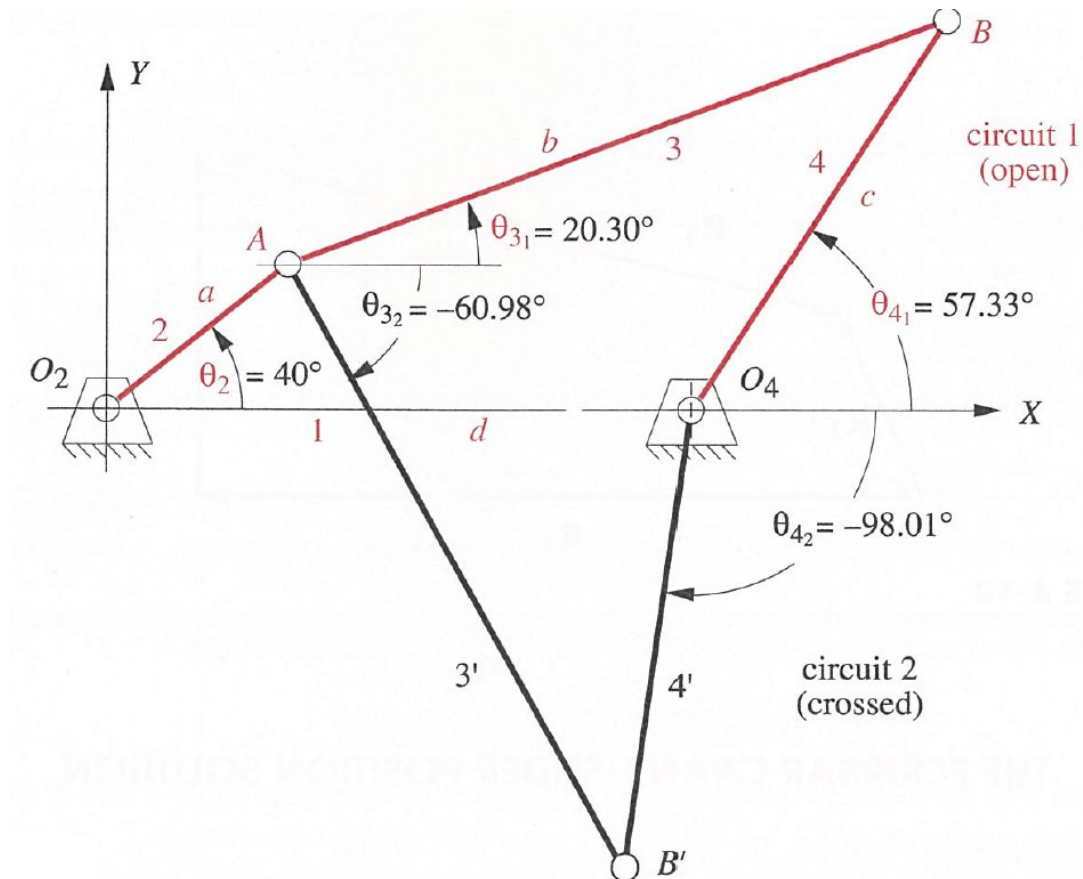
$$\omega_3 = \frac{a\omega_2}{b} \frac{\sin(\theta_4 - \theta_2)}{\sin(\theta_3 - \theta_4)}$$

$$\omega_4 = \frac{a\omega_2}{c} \frac{\sin(\theta_2 - \theta_3)}{\sin(\theta_4 - \theta_3)}$$

$$\underline{V_A} = ja\omega_2 e^{j\theta_2} \quad \underline{V_{BA}} = jb\omega_3 e^{j\theta_3} \quad \underline{V_B} = jc\omega_4 e^{j\theta_4}$$

Example 1

Given a fourbar linkage with the link lengths $L_1 = d = 100$ mm, $L_2 = a = 40$ mm, $L_3 = b = 120$ mm, $L_4 = c = 80$ mm. For $\theta_2 = 40^\circ$ and $\omega_2 = 25$ rad/sec find the values of ω_3 and ω_4 , V_A , V_{BA} , and V_B for the open circuit of the linkage. Use the angles found for the same linkage and position in Example 4-1.



Solution

$$\omega_3 = \frac{a\omega_2}{b} \frac{\sin(\theta_4 - \theta_2)}{\sin(\theta_3 - \theta_4)} = \frac{40(25)}{120} \frac{\sin(57.325^\circ - 40^\circ)}{\sin(20.298^\circ - 57.325^\circ)} = -4.121 \text{ rad/sec}$$

$$\omega_4 = \frac{a\omega_2}{c} \frac{\sin(\theta_2 - \theta_3)}{\sin(\theta_4 - \theta_3)} = \frac{40(25)}{80} \frac{\sin(40^\circ - 20.298^\circ)}{\sin(57.325^\circ - 20.298^\circ)} = 6.998 \text{ rad/sec}$$

$$\begin{aligned} \mathbf{V}_A &= a\omega_2(-\sin\theta_2 + j\cos\theta_2) \\ &= 40(25)(-\sin 40^\circ + j\cos 40^\circ) = -642.79 + j766.04 \end{aligned}$$

$$\mathbf{V}_{Ax} = -642.79; \quad \mathbf{V}_{Ay} = 766.04; \quad \mathbf{V}_{A_{mag}} = 1000 \text{ mm/sec}; \quad \mathbf{V}_{A_{ang}} = 130^\circ$$

$$\begin{aligned} \mathbf{V}_{BA} &= b\omega_3(-\sin\theta_3 + j\cos\theta_3) \\ &= 120(-4.121)(-\sin 20.298^\circ + j20.298^\circ) = 171.55 - j463.80 \end{aligned}$$

$$\mathbf{V}_{BA_x} = 171.55; \quad \mathbf{V}_{BA_y} = -463.80; \quad \mathbf{V}_{BA_{mag}} = 494.51 \text{ mm/sec}; \quad \mathbf{V}_{BA_{ang}} = -69.70^\circ$$

$$\begin{aligned} \mathbf{V}_B &= c\omega_4(-\sin\theta_4 + j\cos\theta_4) \\ &= 80(6.998)(-\sin 57.325^\circ + j\cos 57.325^\circ) = -471.242 + j302.243 \end{aligned}$$

$$\mathbf{V}_{B_x} = -471.242; \quad \mathbf{V}_{B_y} = 302.243; \quad \mathbf{V}_{B_{mag}} = 559.84 \text{ mm/sec}; \quad \mathbf{V}_{B_{ang}} = 147.33^\circ$$

– Crank Slider

$$\mathbf{R}_2 - \mathbf{R}_3 - \mathbf{R}_4 - \mathbf{R}_1 = 0 \iff a e^{j\theta_2} - b e^{j\theta_3} - j c - d = 0$$

Differentiate the above with time and note

$$j a \omega_2 e^{j\theta_2} - j b \omega_3 e^{j\theta_3} - \dot{d} = 0$$

$$\mathbf{V}_A - \mathbf{V}_{AB} - \mathbf{V}_B = 0$$



$$-a \omega_2 \sin \theta_2 + b \omega_3 \sin \theta_3 - \dot{d} = 0$$

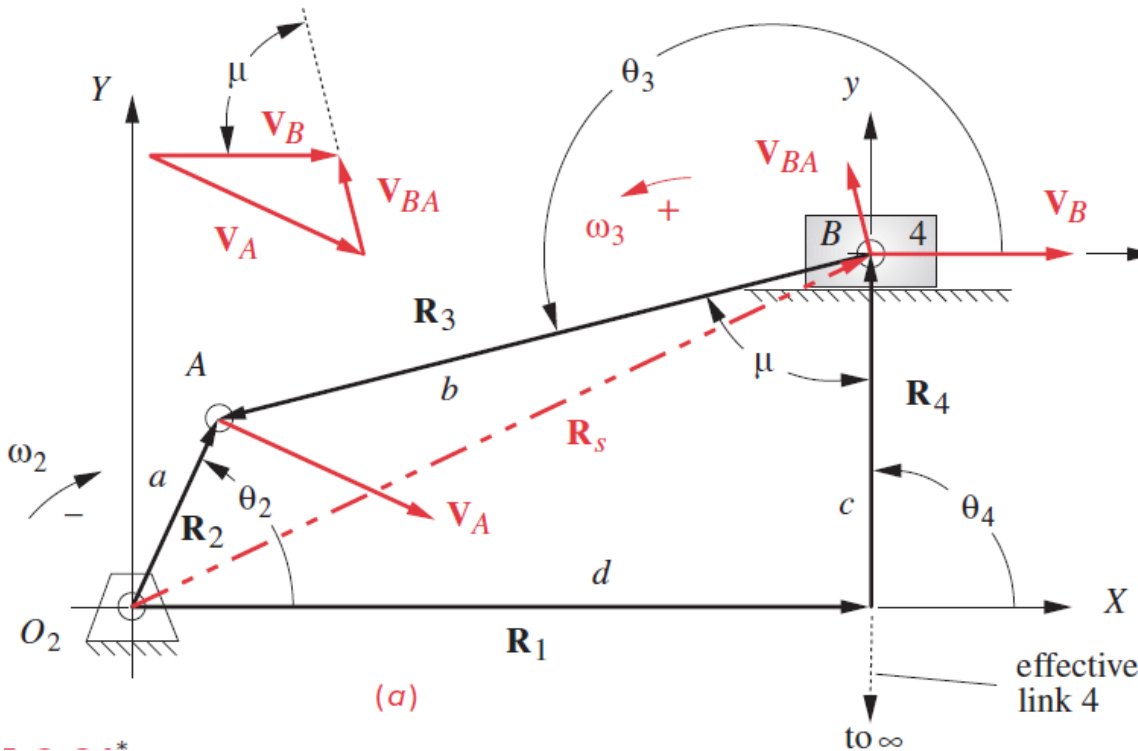
$$a \omega_2 \cos \theta_2 - b \omega_3 \cos \theta_3 = 0$$



$$\omega_3 = \frac{a \cos \theta_2}{b \cos \theta_3} \omega_2$$

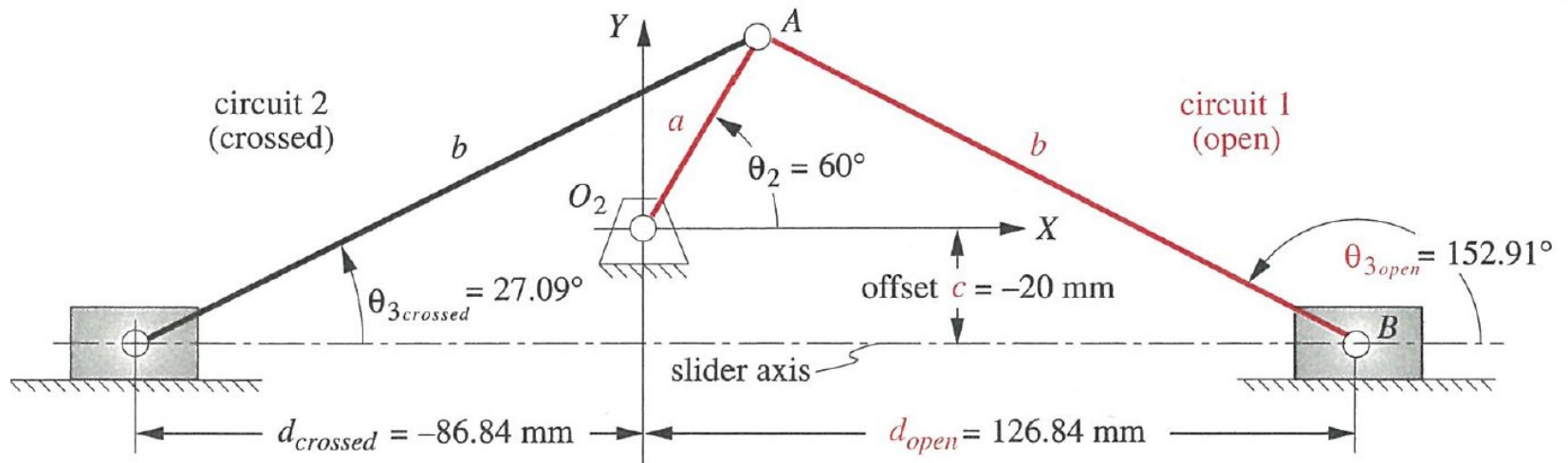
$$\dot{d} = -a \omega_2 \sin \theta_2 + b \omega_3 \sin \theta_3$$

$$\mathbf{V}_A = a \omega_2 (-\sin \theta_2 + j \cos \theta_2) \quad \mathbf{V}_{AB} = b \omega_3 (-\sin \theta_3 + j \cos \theta_3)$$



Example 2

Given a fourbar crank-slider linkage with the link lengths $L_2 = a = 40$ mm, $L_3 = b = 120$ mm, *offset* = $c = -20$ mm. For $\theta_2 = 60^\circ$ and $\omega_2 = -30$ rad/sec, find ω_3 and linear velocities of points A and B for the open circuit. Use the angles and positions found for the same linkage and its link 2 position in Example 4-2.



Solution

$$\omega_3 = \frac{a \cos \theta_2}{b \cos \theta_3} \omega_2 = \frac{40}{120} \frac{\cos 60^\circ}{\cos 152.91^\circ} (-30) = 5.616 \text{ rad/sec}$$

$$\dot{d} = -a \omega_2 \sin \theta_2 + b \omega_3 \sin \theta_3 = -40(-30) \sin 60^\circ + 120(5.616) \sin 152.91^\circ = 1346 \text{ mm/sec}$$

$$\mathbf{V}_A = a \omega_2 (-\sin \theta_2 + j \cos \theta_2) = 40(-30)(-\sin 60^\circ + j \cos 60^\circ) = 1039.23 - j600$$

$$\mathbf{V}_{A_x} = 1039.23; \quad \mathbf{V}_{A_y} = -600; \quad \mathbf{V}_{A_{mag}} = 1200 \text{ mm/sec}; \quad \mathbf{V}_{A_{ang}} = -30^\circ$$

$$\begin{aligned} \mathbf{V}_{AB} &= b \omega_3 (-\sin \theta_3 + j \cos \theta_3) \\ &= 120(5.616)(-\sin 152.91^\circ + j \cos 152.91^\circ) = -306.86 - j600 \end{aligned}$$

$$\mathbf{V}_{BA} = -\mathbf{V}_{AB} = 306.86 + j600$$

$$\mathbf{V}_{BA_x} = 306.86; \quad \mathbf{V}_{BA_y} = 600; \quad \mathbf{V}_{BA_{mag}} = 673.92 \text{ mm/sec}; \quad \mathbf{V}_{BA_{ang}} = 62.91^\circ$$

– Slider Crank

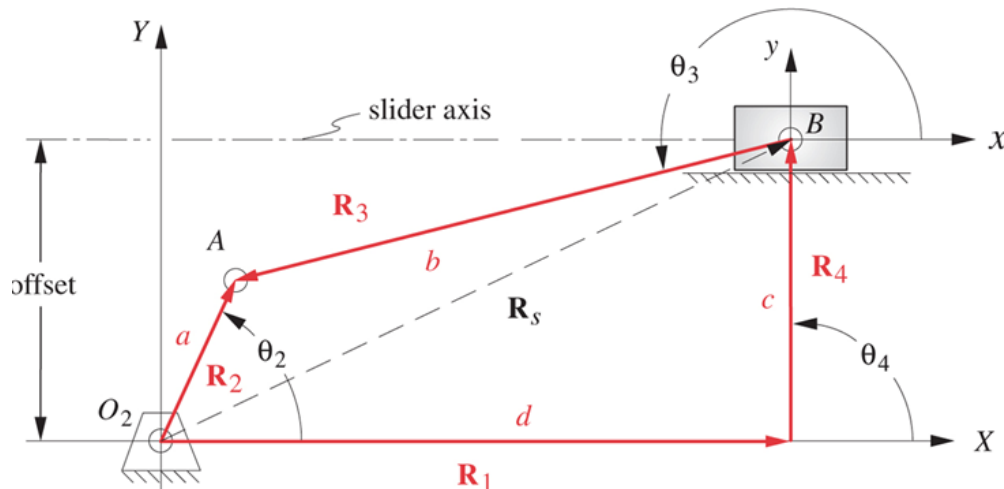
Same velocity equations, but different input/output !

$$\begin{aligned} -a\omega_2 \sin\theta_2 + b\omega_3 \sin\theta_3 - \dot{d} &= 0 \\ a\omega_2 \cos\theta_2 - b\omega_3 \cos\theta_3 &= 0 \end{aligned}$$



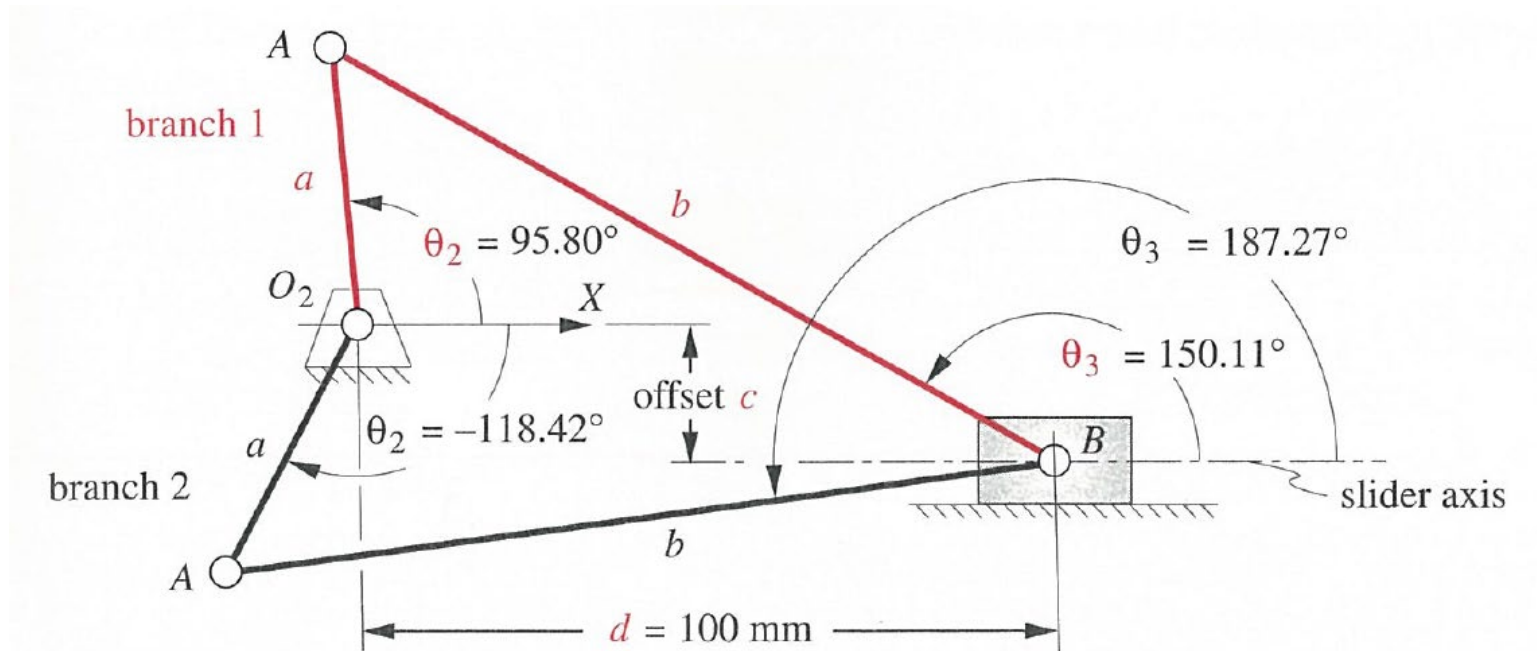
$$\omega_3 = \frac{a\omega_2 \cos\theta_2}{b \cos\theta_3}$$

$$\omega_2 = \frac{\dot{d} \cos\theta_3}{a(\cos\theta_2 \sin\theta_3 - \sin\theta_2 \cos\theta_3)}$$



Example 3

Given a fourbar slider-crank linkage with the link lengths $L_2 = a = 40$ mm, $L_3 = b = 120$ mm, *offset* = $c = -20$ mm. For $d = 100$ mm and $\dot{d} = 1200$ mm/sec, find ω_2 and ω_3 for both branches of one circuit of the linkage. Use the angles found for the same linkage in Example 4-3.



Solution

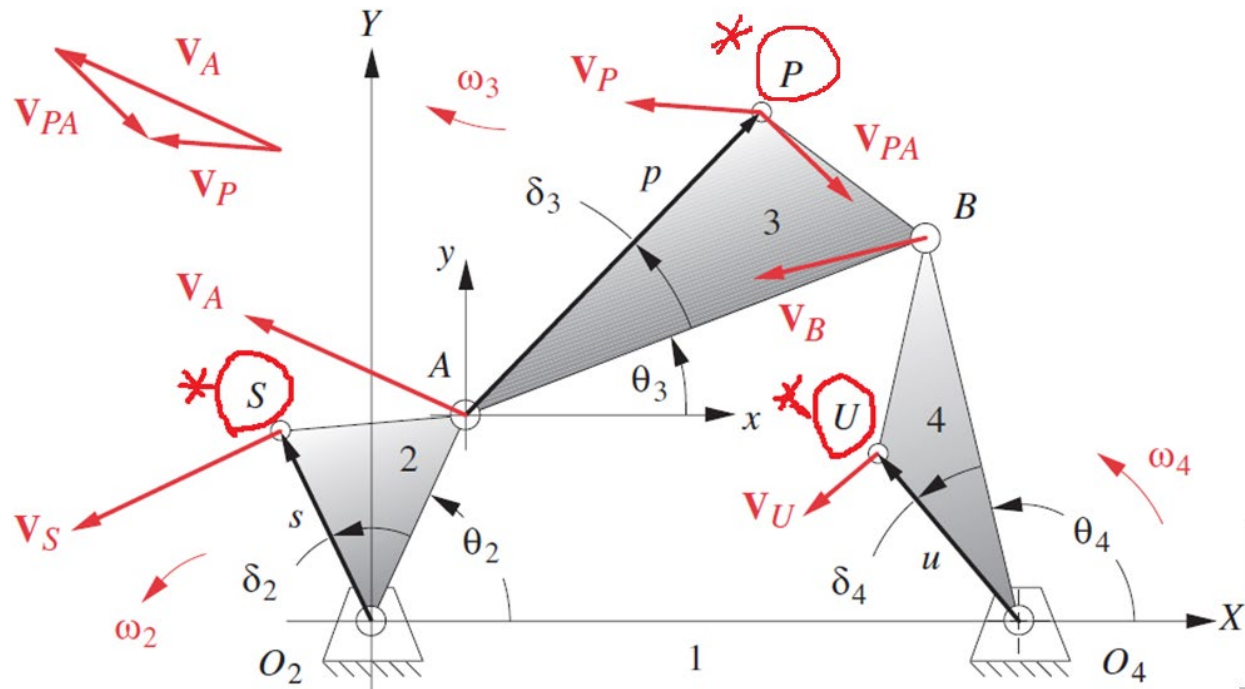
$$\omega_{2_1} = \frac{\dot{d} \cos \theta_{3_1}}{a \left(\cos \theta_{2_1} \sin \theta_{3_1} - \sin \theta_{2_1} \cos \theta_{3_1} \right)} = -32.023 \text{ rad/sec}$$

$$\omega_{3_1} = \frac{a \omega_{2_1} \cos \theta_{2_1}}{b \cos \theta_{3_1}} = -1.244 \text{ rad/sec}$$

$$\omega_{2_2} = \frac{\dot{d} \cos \theta_{3_2}}{a \left(\cos \theta_{2_2} \sin \theta_{3_2} - \sin \theta_{2_2} \cos \theta_{3_2} \right)} = 36.639 \text{ rad/sec}$$

$$\omega_{3_2} = \frac{a \omega_{2_2} \cos \theta_{2_2}}{b \cos \theta_{3_2}} = 5.859 \text{ rad/sec}$$

- Velocity of any point in a linkage



$$\mathbf{R}_{SO_2} = \mathbf{R}_S = se^{j(\theta_2 + \delta_2)} = s[\cos(\theta_2 + \delta_2) + j\sin(\theta_2 + \delta_2)]$$

$$\mathbf{R}_{UO_4} = ue^{j(\theta_4 + \delta_4)} = u[\cos(\theta_4 + \delta_4) + j\sin(\theta_4 + \delta_4)]$$

$$\mathbf{V}_S = jse^{j(\theta_2 + \delta_2)}\omega_2 = s\omega_2[-\sin(\theta_2 + \delta_2) + j\cos(\theta_2 + \delta_2)]$$

$$\mathbf{V}_U = jue^{j(\theta_4 + \delta_4)}\omega_4 = u\omega_4[-\sin(\theta_4 + \delta_4) + j\cos(\theta_4 + \delta_4)]$$

$$\mathbf{R}_{PA} = pe^{j(\theta_3 + \delta_3)}$$

$$\mathbf{V}_{PA} = jpe^{j(\theta_3 + \delta_3)}\omega_3 :$$

$$\mathbf{V}_P = \mathbf{V}_A + \mathbf{V}_{PA}$$